

Jack Patchell
503 857 3879 | jackpatchell222@gmail.com | <https://jpatch22.github.io/>

Education

University of British Columbia, Vancouver, BC BASc Engineering Physics, Minor Commerce GPA: 4.33/4.33 Honors: Trek Scholarship (Top 5% Faculty), Deans List Languages: English (Native), Mandarin (Intermediate)	Sept. 2019-May 2025
--	----------------------------

Experience

Cloud Software Engineering Intern – Rivian – Palo Alto, CA	May 2024-Aug.2024
---	--------------------------

- Built frontend component of features in Andriod.
- Worked on cloud development with Spring, Kotlin and Redis and monitored deployment using Datadog.
- Built a React app to show detailed logging information including vehicle routes to enhance debugging.

Software Engineering Intern – Rivian – Palo Alto, CA	May 2023-Sep.2023
---	--------------------------

- Led a full-stack project that integrated Google's ETA estimations, resulting in a reduction of cloud API calls by 50% during search, and elevating the service.
- Contributions to add-stop project, including refactor of drawing routes, camera behavior and locations pins. Led a refactor towards reactive programming model/Flow driven style for routes, map pins, and camera behavior.

Software Engineering Intern – Rivian – Vancouver, BC	May 2022-Dec. 2022
---	---------------------------

- Led vehicle implementation of a feature enabling seamless address sharing from mobile to vehicle, enhancing user convenience through robust Kotlin and Java implementation.
- Resolved 30+ bugs and introduced enhancements to the Navigation App, including entity parsing, data formatting, and dynamic icon construction, contributing to OTA updates across 25,000+ vehicles.

Computer Engineering Intern – Svante – Burnaby, BC	Jan. 2021-Aug. 2021
---	----------------------------

- Conducted time-series data analysis on a carbon capture pilot unit using SEEQ, applying statistical methods to calculate real-time KPIs, aiding strategic decision-making for business stakeholders.

Selected Technical Projects

Model Deployment – Fast inference on FPGA DPU
--

- Led custom model deployment onto the Xilinx VCK5000, explored hardware acceleration, power usage quantization strategies and model compression.
- Built demo application to run the model on the CPU and Xilinx VCK5000, compare accuracy and show results inference speed.

Machine Learning — Bat Vision – Capstone Research Project
--

- Led machine learning effort to convert audio signal inputs into 3D spatial images implementing a convolution neural network with TensorFlow.
- Leveraged Wandb for rigorous tracking and dashboard analytics, streamlining experiment record-keeping and enhancing model performance tuning.

Machine Learning – Autonomous Vehicle
--

- Robot in action: https://www.youtube.com/watch?v=IXj0uXl4V1I&ab_channel=JackPatchell
- Designed an autonomous car solution to follow traffic regulations in ROS Gazebo simulation.
- Implemented a Convolutional Neural Network using Keras for accurate license plate scanning.

Firmware Development – Robotics Competition
--

- Led firmware development in C, programming a blue pill microcontroller to manage robot navigation, and pick up cans to place them in a container.

Skills

Kotlin, Java, Android, OOP, Reactive Programming, Machine Learning, Fast AI, Hugging Face, TensorFlow, Keras, Python, Redis, Spring, Typescript/Javascript, React